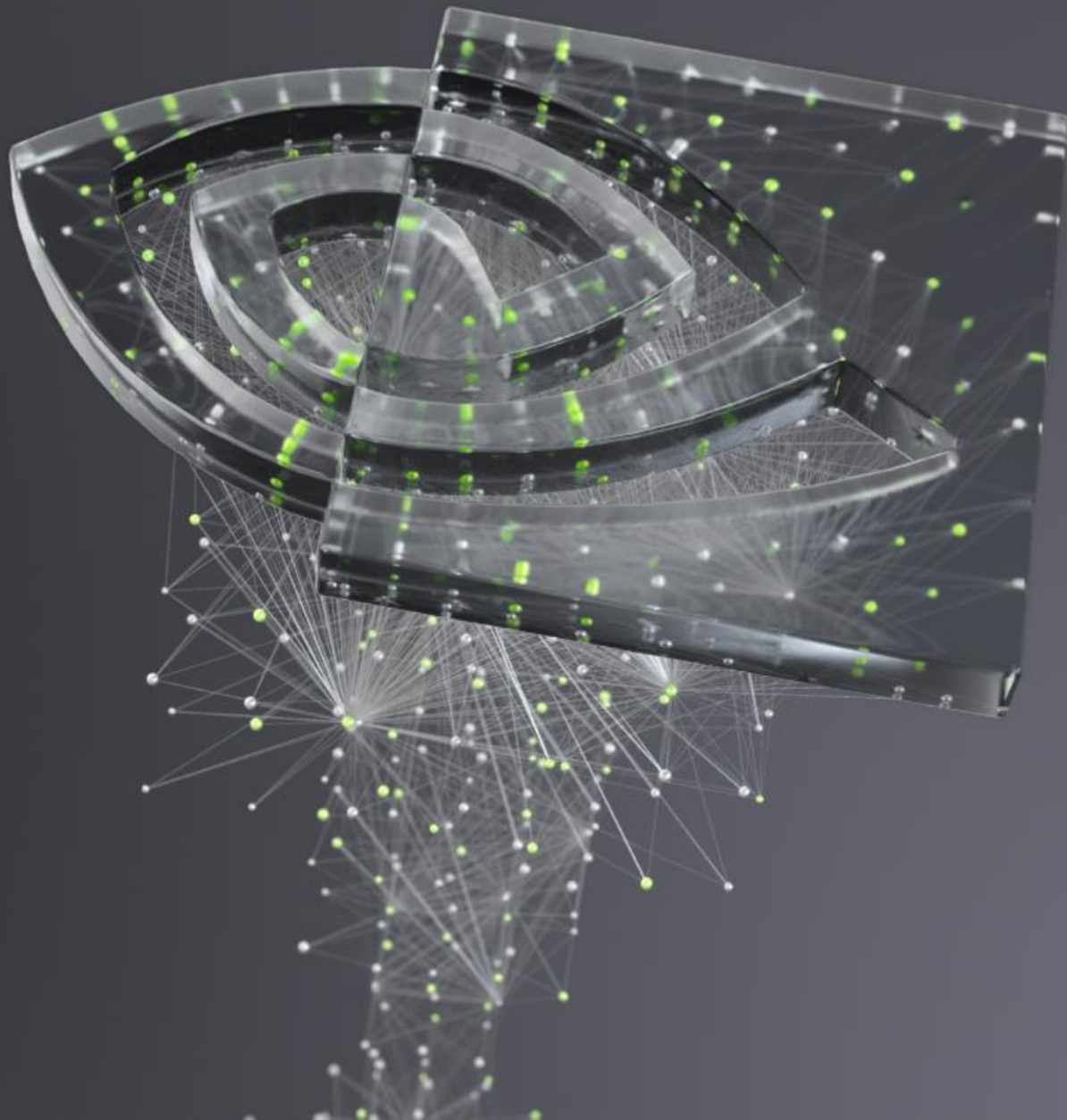




DEEP
LEARNING
INSTITUTE

딥러닝의 기초

6부: 고급 아키텍처



목차

1부: 딥러닝 소개

2부: 뉴럴 네트워크의 트레이닝 방식

3부: CNN(Convolutional Neural Network)

4부: 데이터 증강 및 배포

5부: 사전 트레이닝된 모델

6부: 고급 아키텍처

목차 – 6부

- 다음 단계
- 자연어 처리
- RNN(Recurrent Neural Network)
- 기타 아키텍처
- 결론



다음 단계

AI 분야



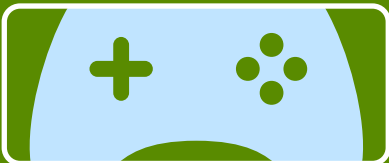
컴퓨터 비전

- 검안법(Optomerty)



자연어 처리

- 언어학



강화 학습

- 게임 이론
- 심리학



이상 탐지

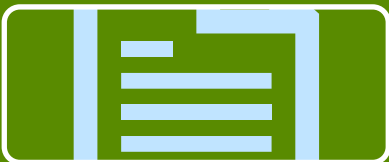
- 보안
- 의학

AI 분야



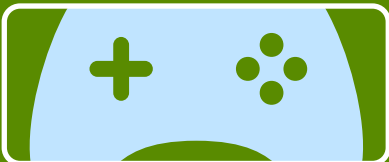
컴퓨터 비전

- 검안법(Optomerty)



자연어 처리

- 언어학



강화 학습

- 게임 이론
- 심리학



이상 탐지

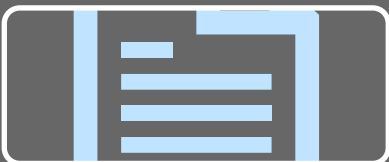
- 보안
- 의학

AI 분야



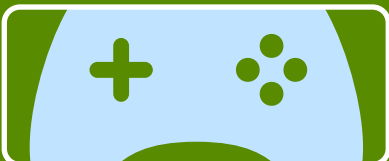
컴퓨터 비전

- 검안법(Optomerty)



자연어 처리

- 언어학



강화 학습

- 게임 이론
- 심리학



이상 탐지

- 보안
- 의학



자연어 처리
NATURAL LANGUAGE PROCESSING

단어에서 숫자로

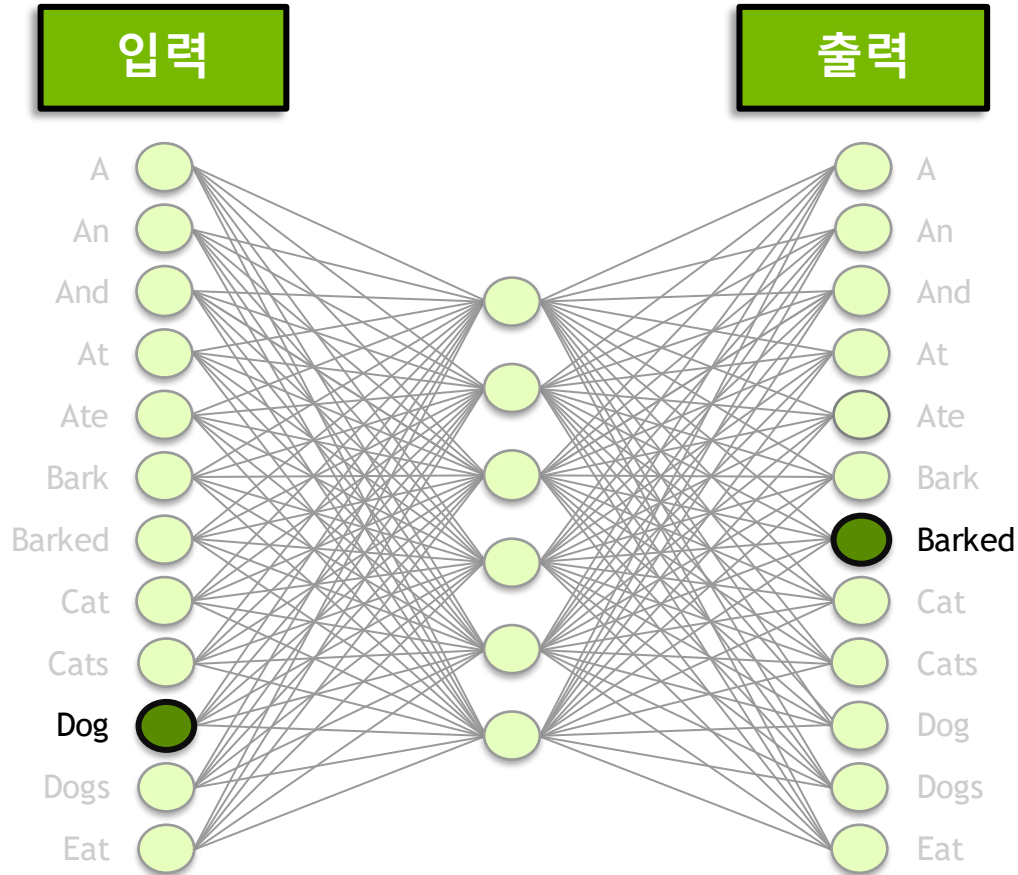
“A dog barked at a cat.”

[1, 10, 7, 4, 1, 8]

사전

- | | |
|-----------|----------|
| 1. A | 8. CAT |
| 2. AN | 9. CATS |
| 3. AND | 10. DOG |
| 4. AT | 11. DOGS |
| 5. ATE | 12. EAT |
| 6. BARK | |
| 7. BARKED | |

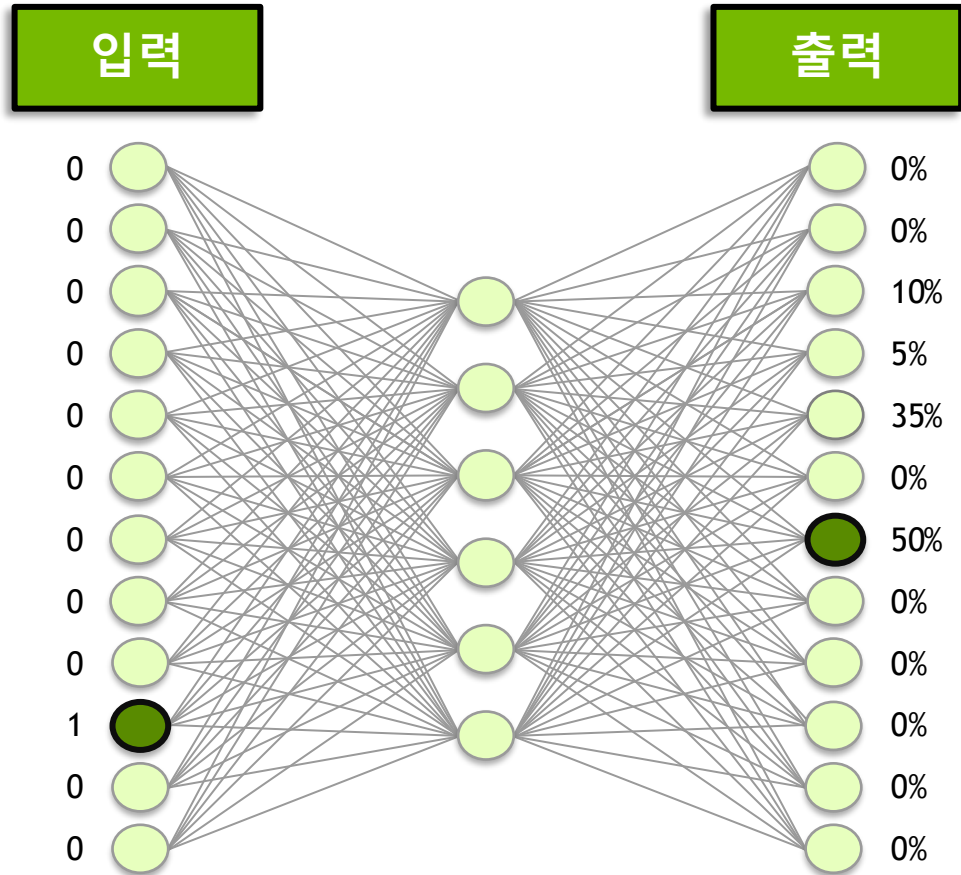
단어에서 숫자로



사전

- | | |
|-----------|----------|
| 1. A | 8. CAT |
| 2. AN | 9. CATS |
| 3. AND | 10. DOG |
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| 6. BARK | |
| 7. BARKED | |

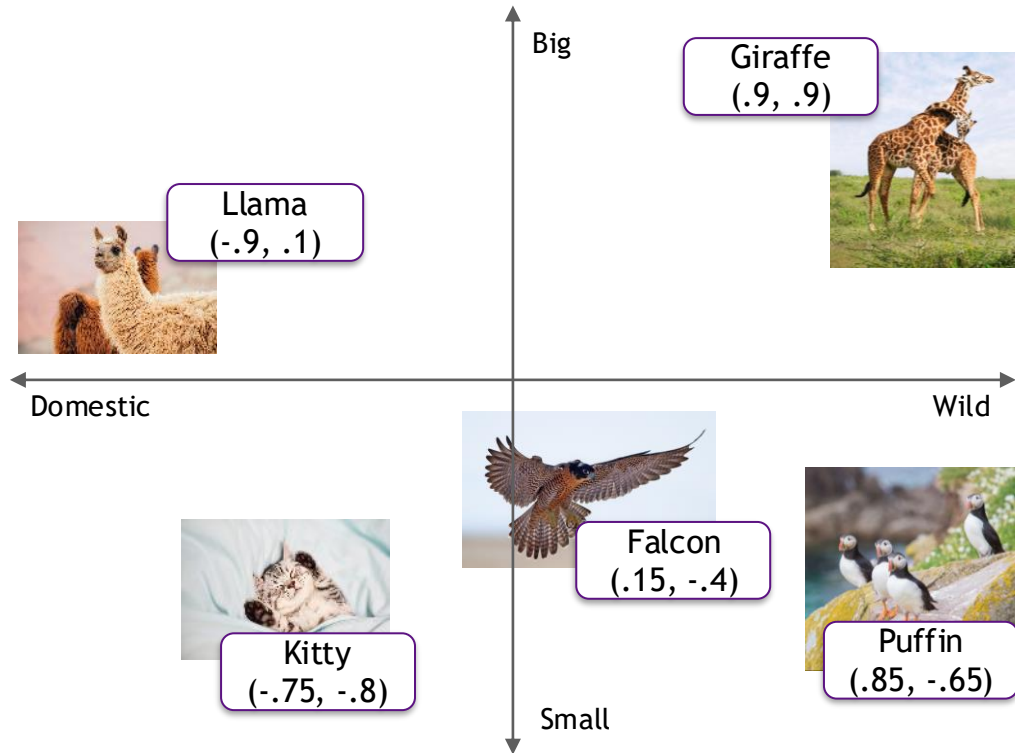
단어에서 숫자로



사전

- | | |
|-----------|----------|
| 1. A | 8. CAT |
| 2. AN | 9. CATS |
| 3. AND | 10. DOG |
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| 5. ATE | 12. EAT |
| 6. BARK | |
| 7. BARKED | |

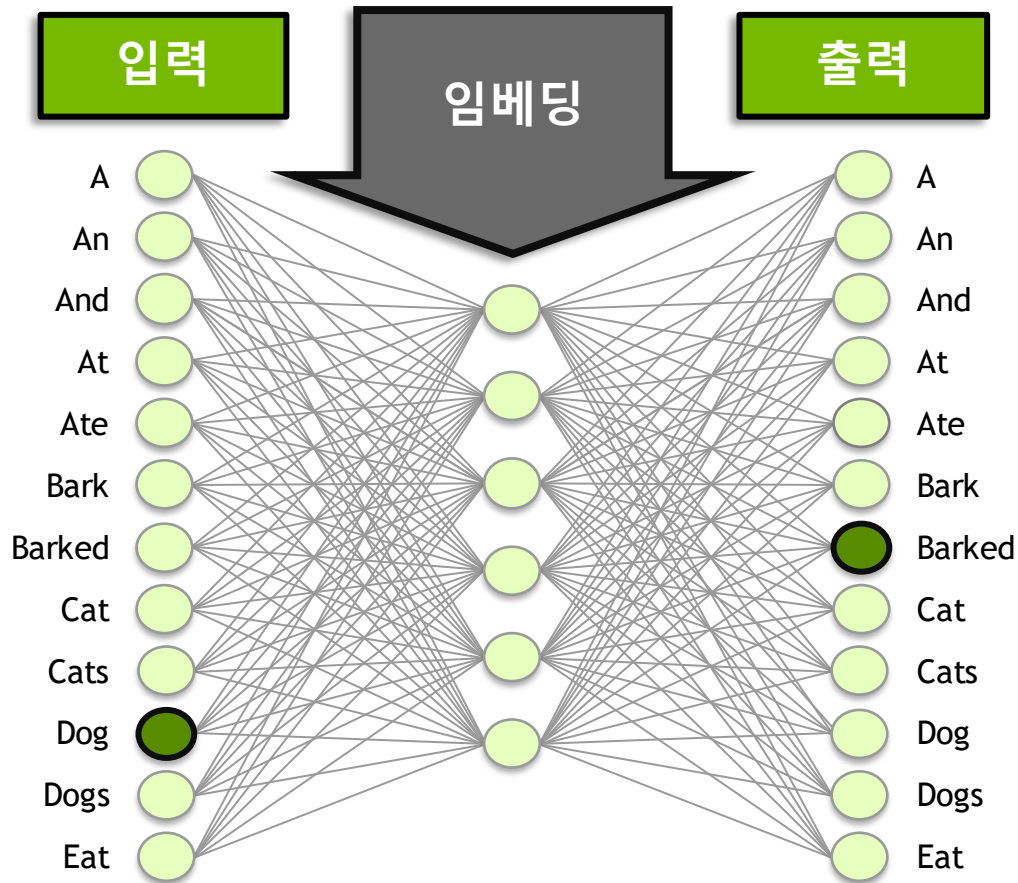
단어에서 숫자로



더 큰 사전

1.	A	31.	ATE	61.	CATS
2.	AN	32.	BARK	62.	DOG
3.	AND	33.	BARKED	63.	DOGS
4.	AT	34.	CAT	64.	EAT
5.	ATE	35.	CATS	65.	EATEN
6.	BARK	36.	DOG	66.	A
7.	BARKED	37.	DOGS	67.	AN
8.	CAT	38.	EAT	68.	AND
9.	CATS	39.	EATEN	69.	AT
10.	DOG	40.	A	70.	ATE
11.	DOGS	41.	AN	71.	BARK
12.	EAT	42.	AND	72.	BARKED
13.	EATEN	43.	AT	73.	CAT
14.	A	44.	ATE	74.	CATS
15.	AN	45.	BARK	75.	DOG
16.	AND	46.	BARKED	76.	DOGS
17.	AT	47.	CAT	77.	EAT
18.	ATE	48.	CATS	78.	EATEN
19.	BARK	49.	DOG	79.	...
20.	BARKED	50.	DOGS	80.	...
21.	CAT	51.	EAT	81.	...
22.	CATS	52.	EATEN	82.	...
23.	DOG	53.	A		
24.	DOGS	54.	AN		
25.	EAT	55.	AND		
26.	EATEN	56.	AT		
27.	A	57.	ATE		
28.	AN	58.	BARK		
29.	AND	59.	BARKED		
30.	AT	60.	CAT		

단어에서 숫자로



사전

- | | |
|-----------|----------|
| 1. A | 8. CAT |
| 2. AN | 9. CATS |
| 3. AND | 10. DOG |
| 4. AT | 11. DOGS |
| 5. ATE | 12. EAT |
| 6. BARK | |
| 7. BARKED | |



RNN(RECURRENT NEURAL
NETWORK)

RNN(RECURRENT NEURAL NETWORK)

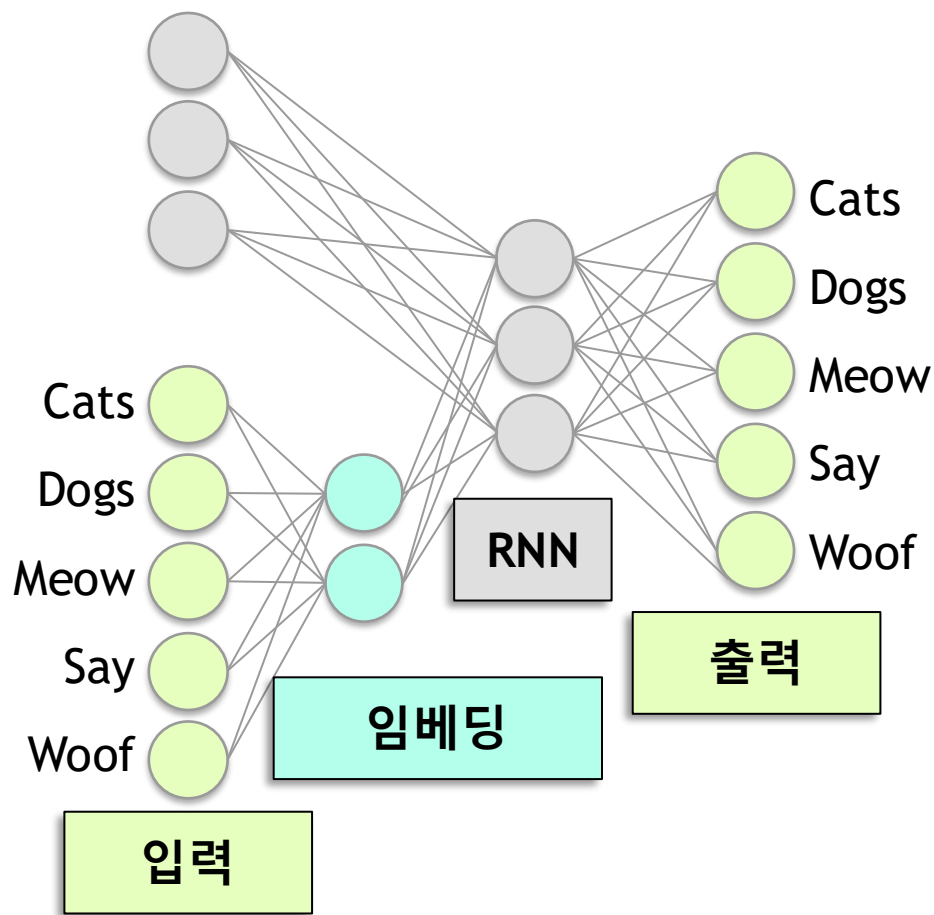
“Cats say ____.”

“Dogs say ____.”

사전

1. CATS
2. DOGS
3. MEOW
4. SAY
5. WOOF

RNN(RECURRENT NEURAL NETWORK)



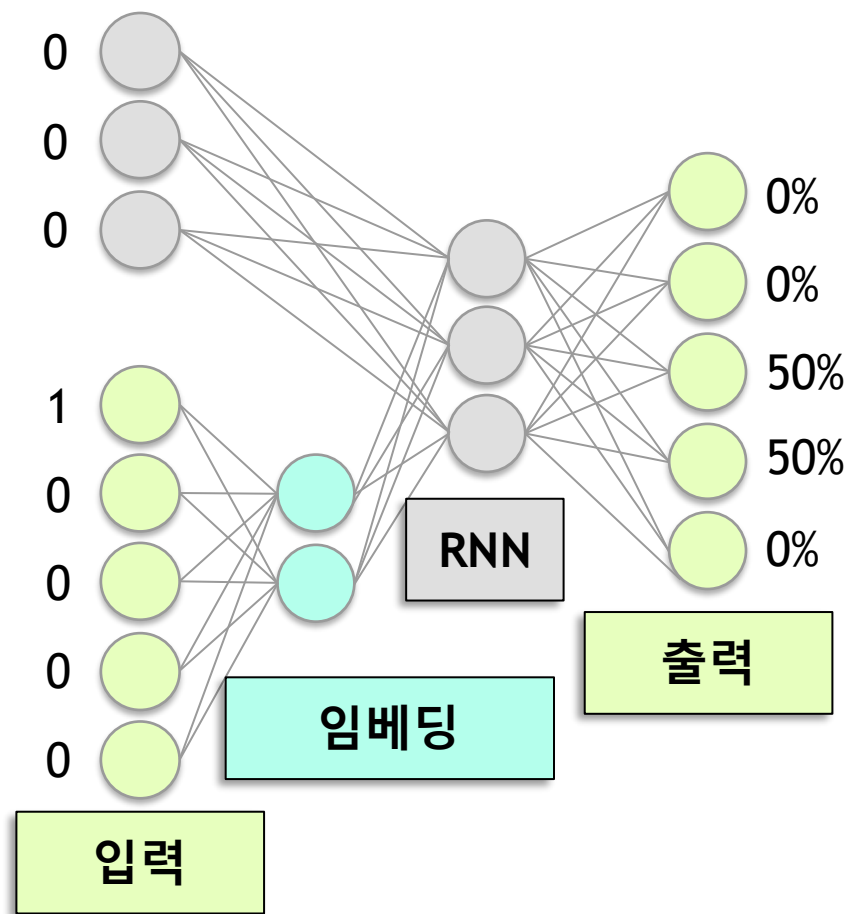
“Cats say ____.”

“Dogs say ____.”

사전

1. CATS
2. DOGS
3. MEOW
4. SAY
5. WOOF

RNN(RECURRENT NEURAL NETWORK)



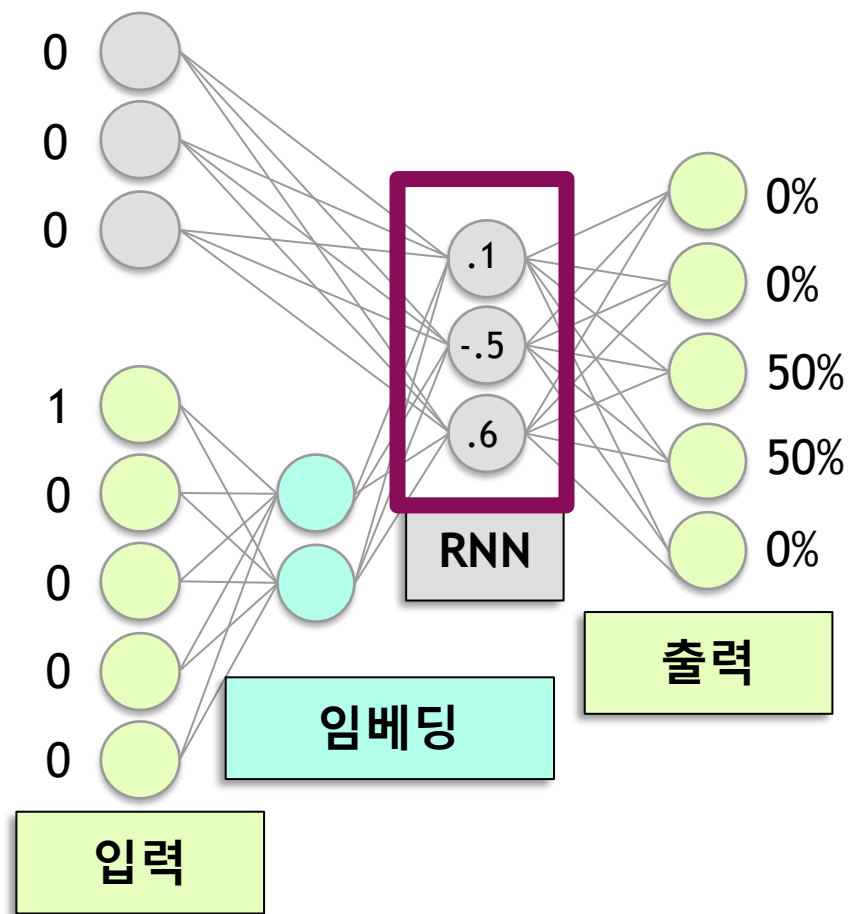
“Cats say ____.”

“Dogs say ____.”

사전

1. CATS
2. DOGS
3. MEOW
4. SAY
5. WOOF

RNN(RECURRENT NEURAL NETWORK)



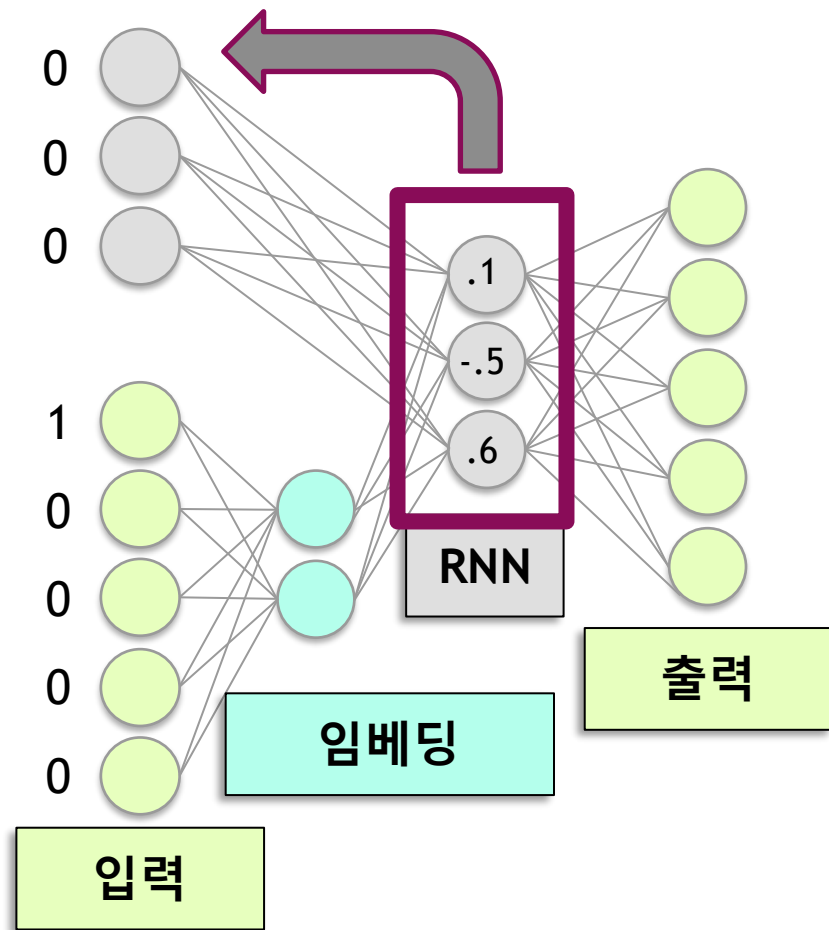
“Cats say ____.”

“Dogs say ____.”

사전

1. CATS
2. DOGS
3. MEOW
4. SAY
5. WOOF

RNN(RECURRENT NEURAL NETWORK)



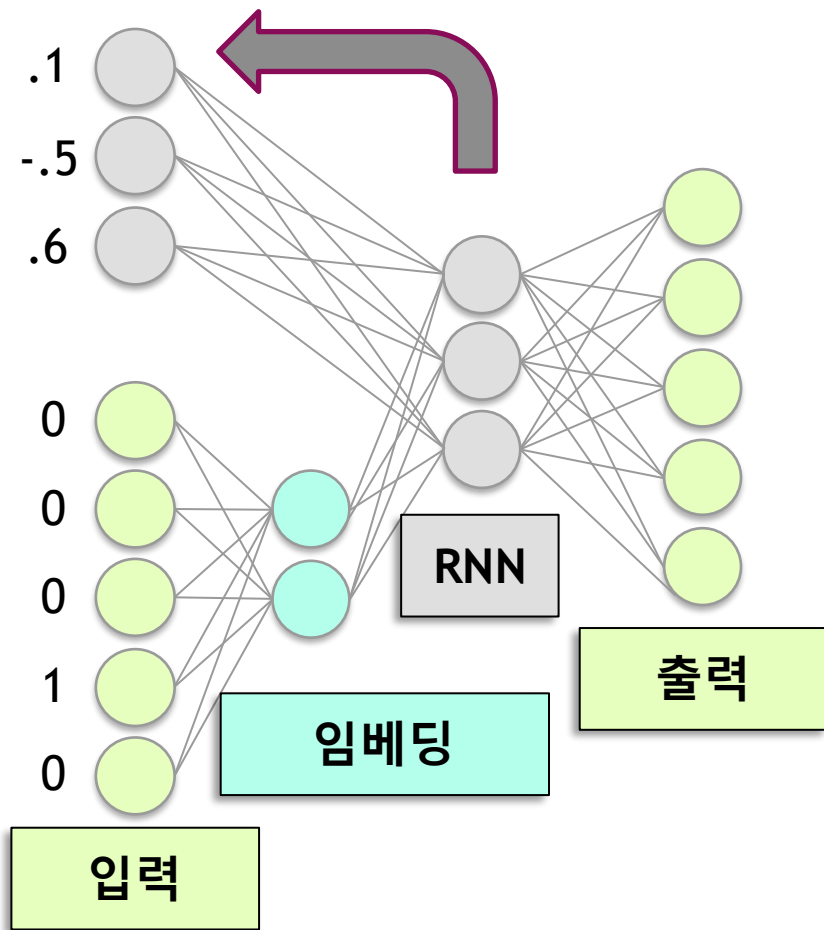
“Cats say ____.”

“Dogs say ____.”

사전

1. CATS
2. DOGS
3. MEOW
4. SAY
5. WOOF

RNN(RECURRENT NEURAL NETWORK)



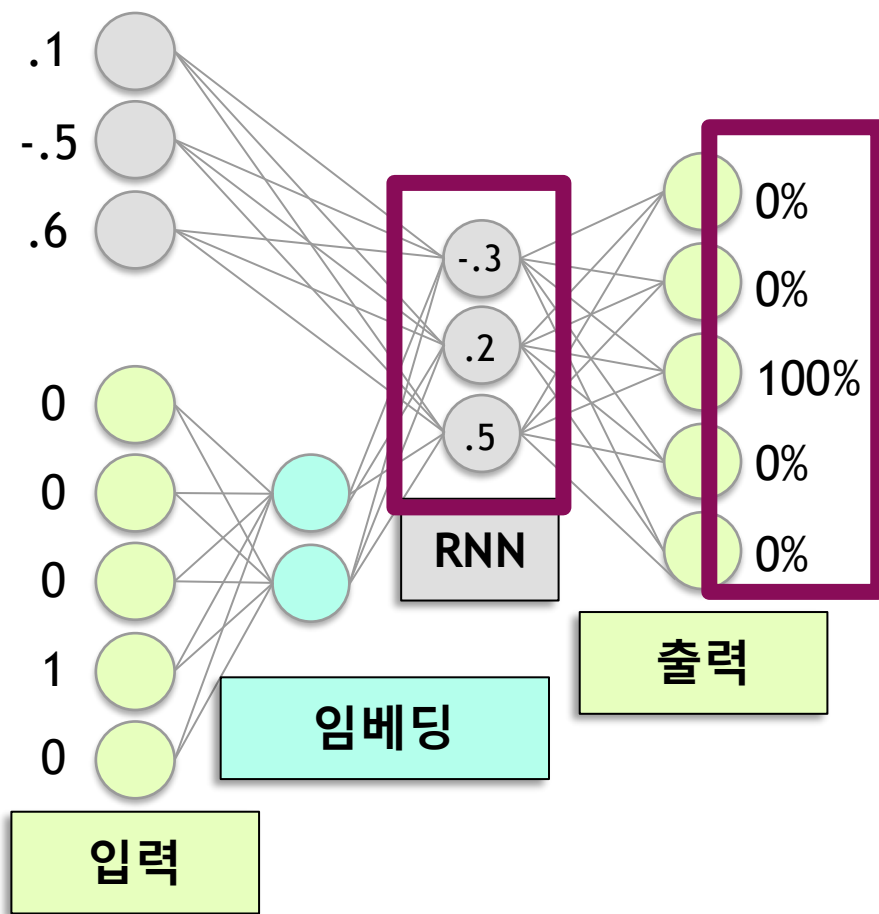
“Cats say ____.”

“Dogs say ____.”

사전

1. CATS
2. DOGS
3. MEOW
4. SAY
5. WOOF

RNN(RECURRENT NEURAL NETWORK)



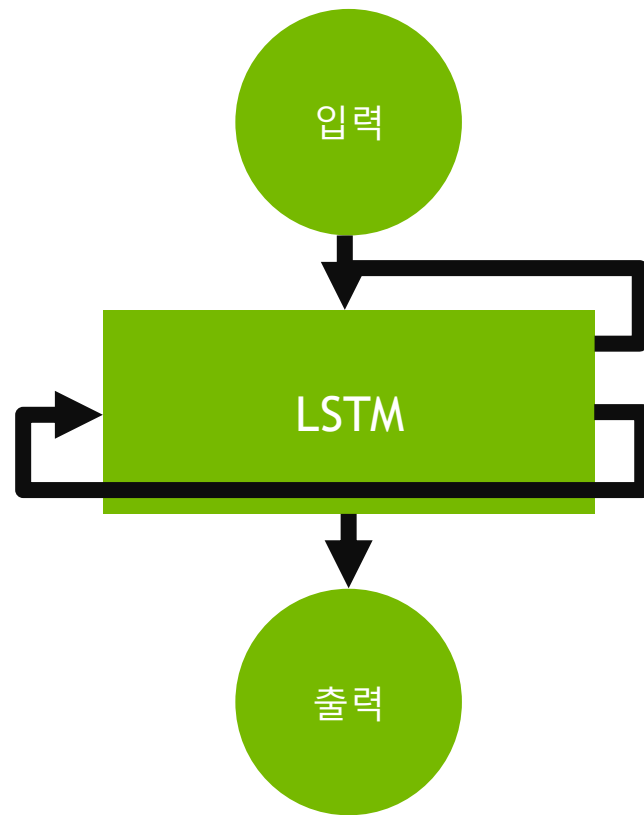
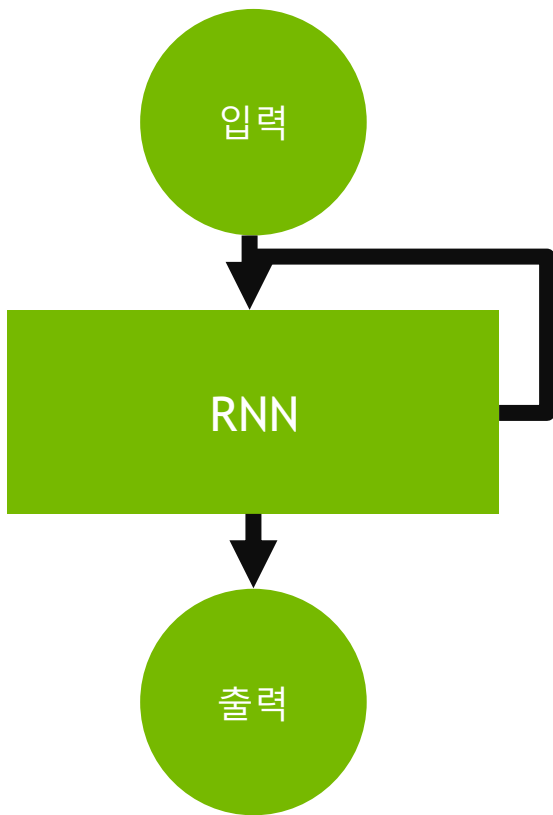
“Cats say ____.”

“Dogs say ____.”

사전

1. CATS
2. DOGS
3. MEOW
4. SAY
5. WOOF

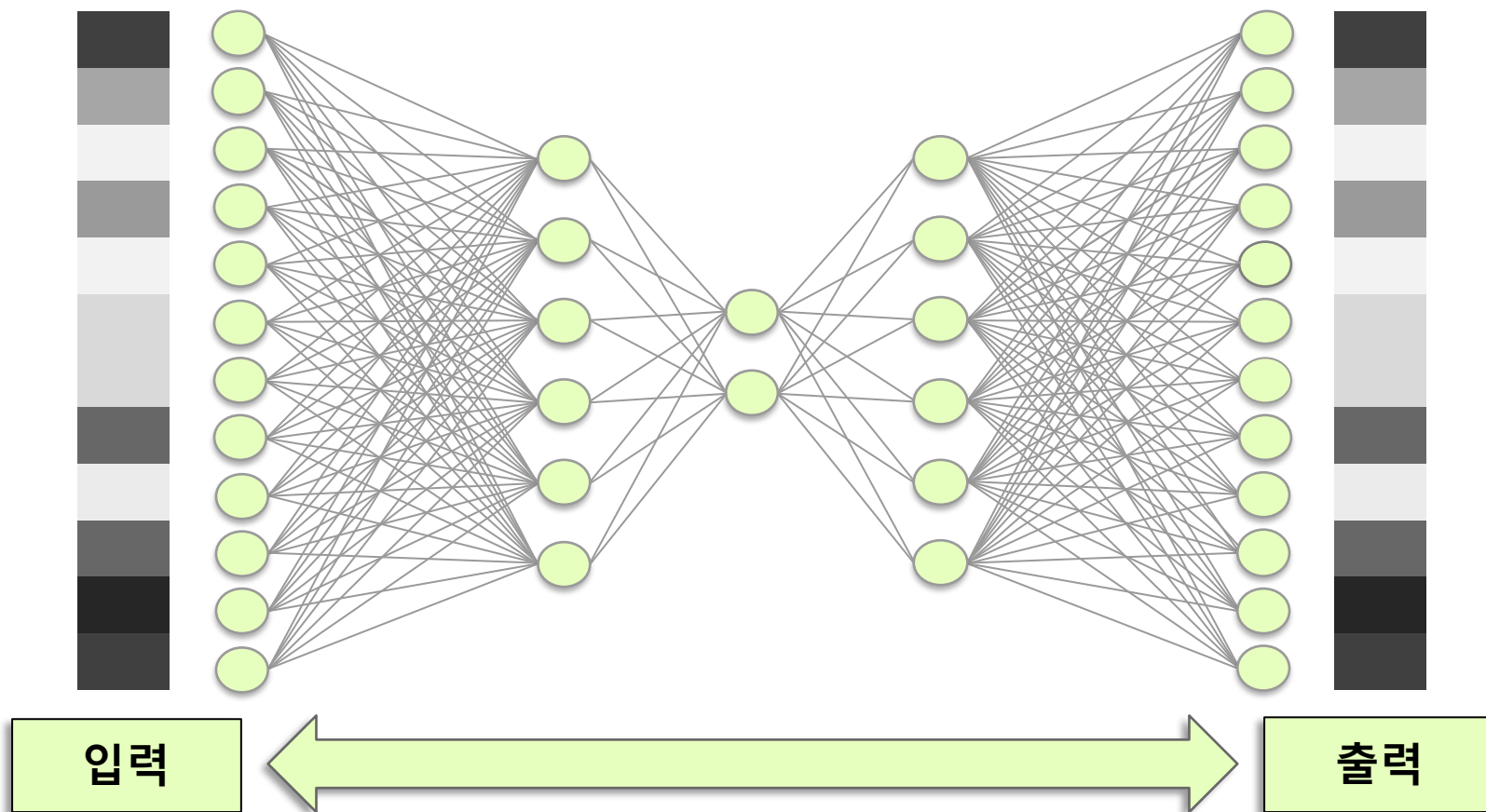
RNN(RECURRENT NEURAL NETWORK)



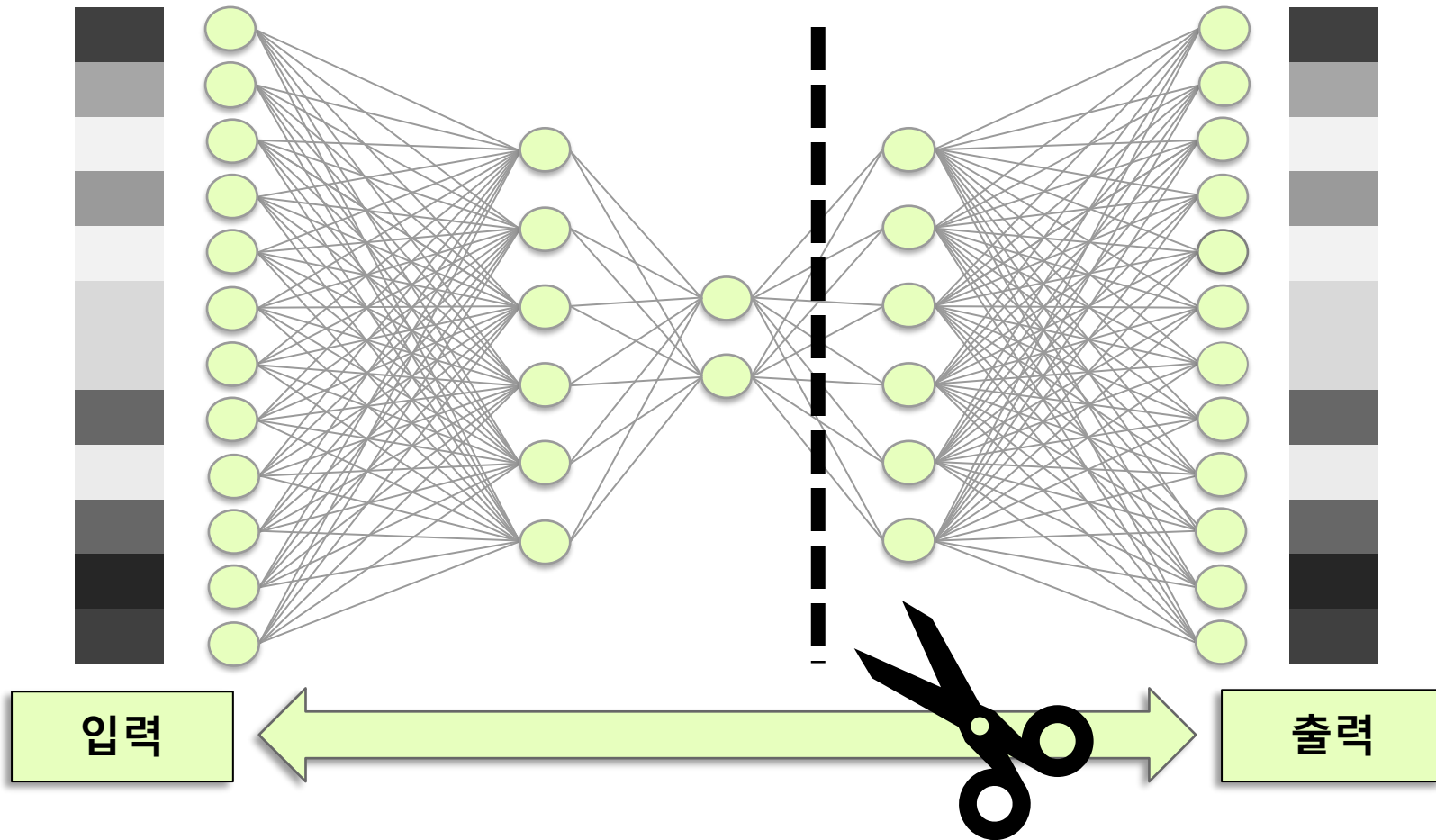


기타 아키텍처 OTHER ARCHITECTURES

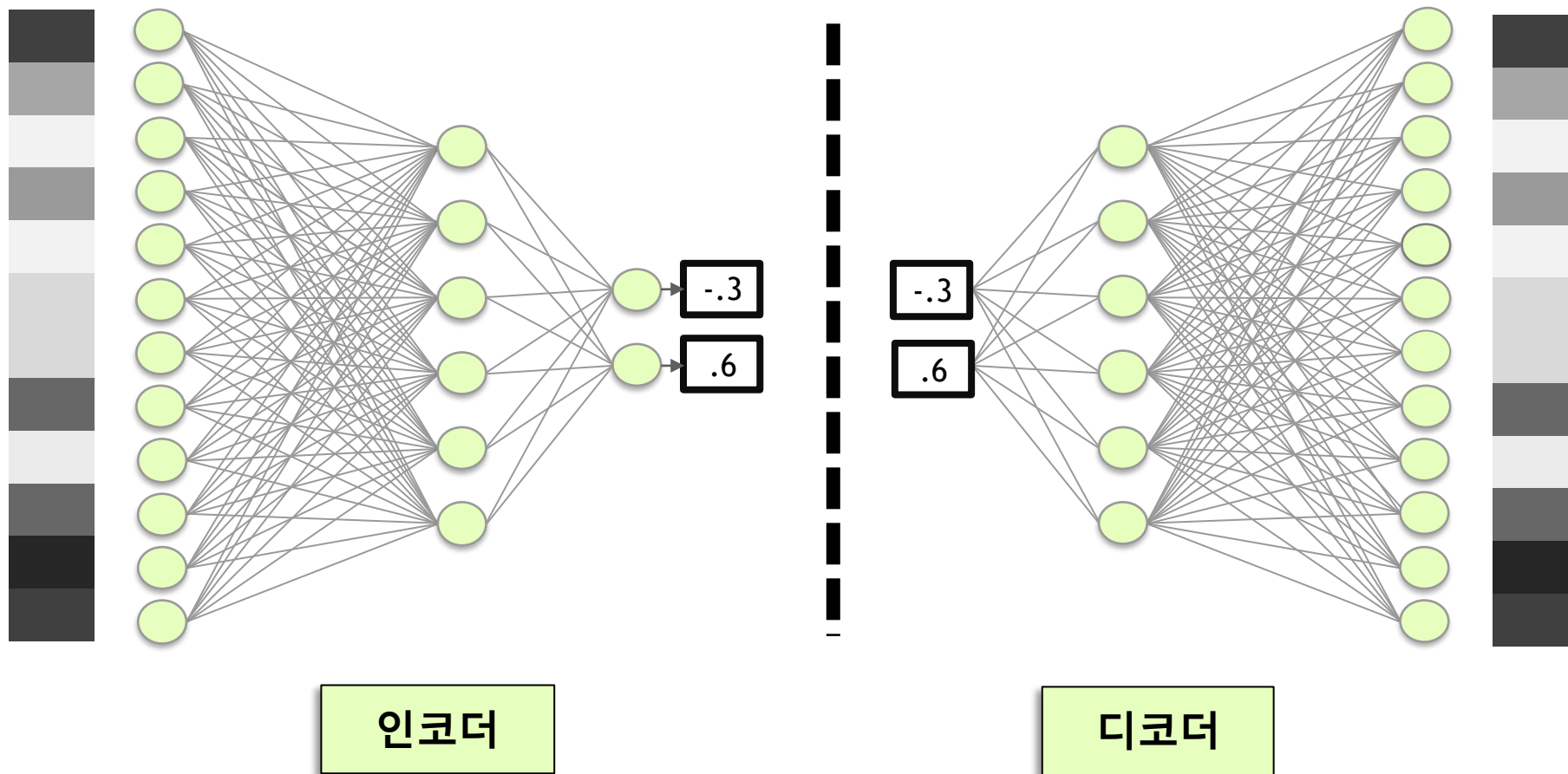
오토인코더(AUTO ENCODER)



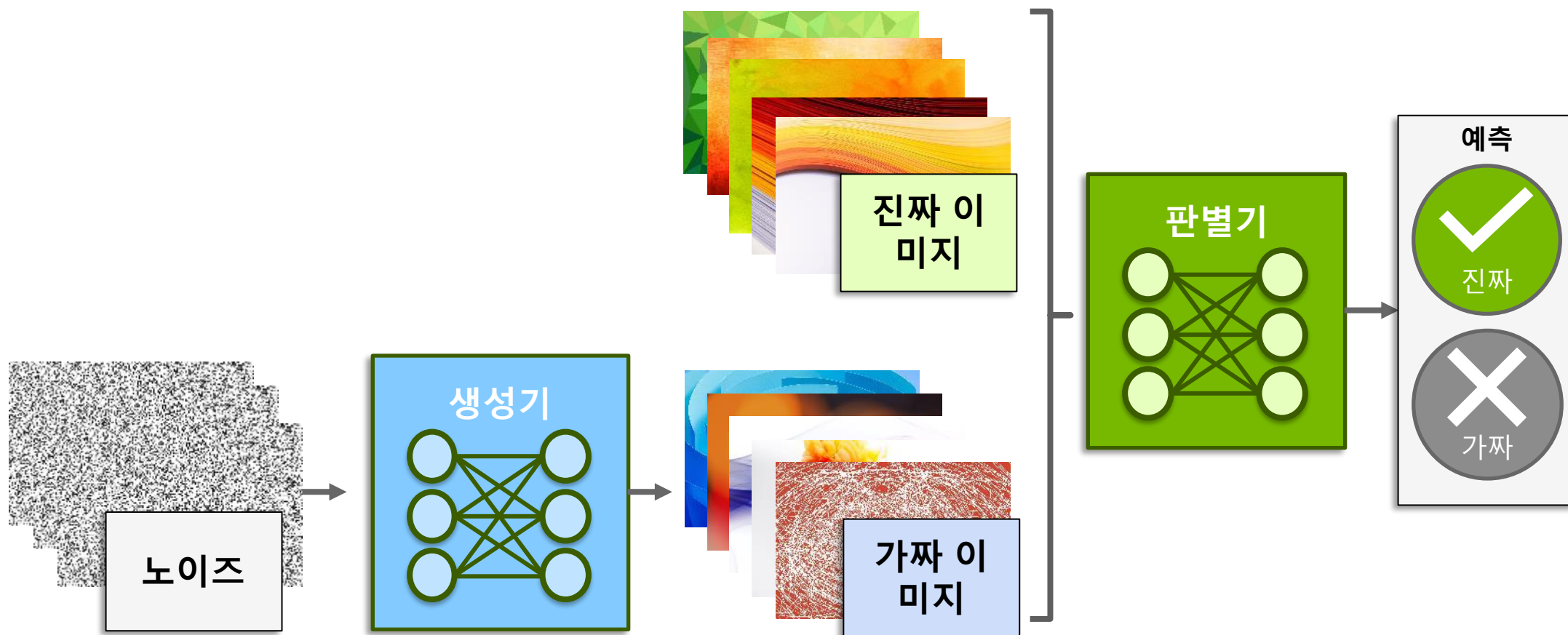
오토인코더(AUTO ENCODER)



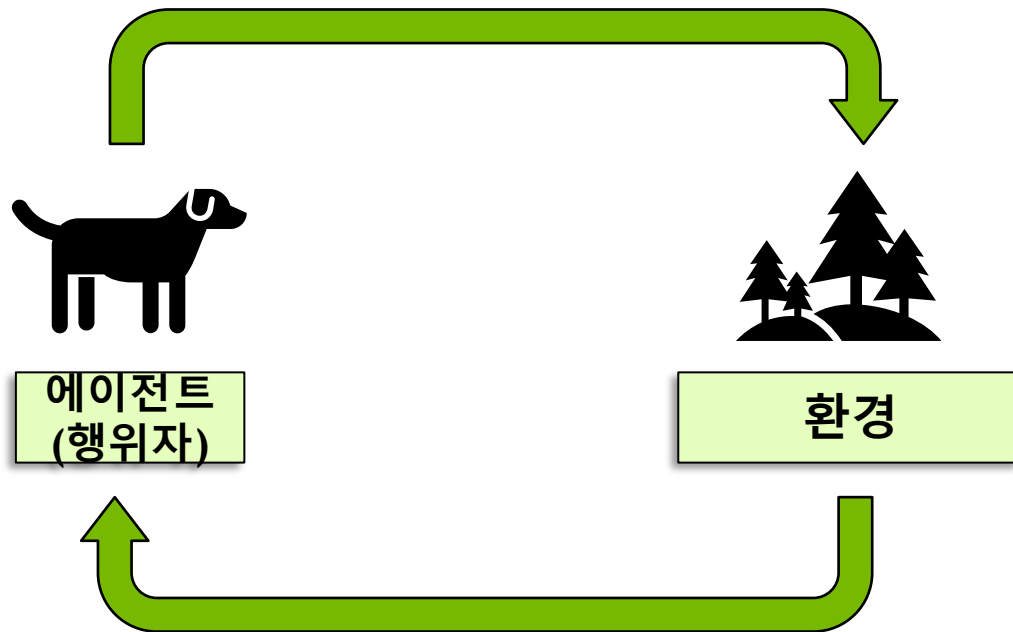
오토인코더(AUTO ENCODER)



GAN(Generative Adversarial Networks)



강화 학습 (REINFORCEMENT LEARNING)





다음 단계

NGC 컨테이너

방대한 범위

- 다양한 범위의 워크로드 및 산업별 사용 사례

최적화

- DL 컨테이너가 매월 업데이트됨
- 최신 기능과 탁월한 성능 보유

보안 및 안정성

- 취약성 및 암호에 대해 검사됨
- 워크스테이션, 서버 및 클라우드 인스턴스에서 테스트됨

확장성

- 멀티 GPU 및 멀티 노드 시스템 지원

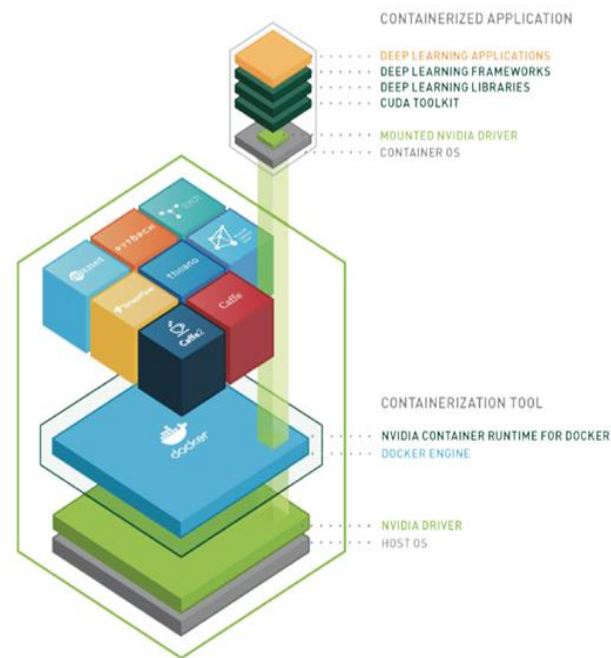
엔터프라이즈 및 HPC용으로 설계됨

- Docker, 특이성(Singularity) 및 기타 런타임 지원

어디서나 실행가능

- 베어메탈, VM, Kubernetes
- X86, ARM, POWER
- 멀티 클라우드, 온-프레미스, 하이브리드, 엣지

NGC 딥러닝 컨테이너



CONVERSATIONAL AI



Riva

HEALTHCARE



CLARA

SMART CITIES



DEEPSTREAM &
SMART PARKING

TELECOM



AERIAL

AUTONOMOUS DRIVING



DRIVE

ROBOTICS



ISAAC

HPC



HPC SDK

[NGC 컨테이너에 대해 자세히 알아보기](#)

이 수업의 다음 단계

Catalog: Containers / Containers: nvidia:dlf-fundamentals

DLF Deep Learning Fundamentals Course -...

Publisher	Built By	Latest Tag	Modified	Size
NVIDIA	NVIDIA	v0.0.1	October 27, 2020	4.19 GB

Multinode Support	Multi-Arch Support
No	

Description

Base environment used in the NVIDIA Deep Learning Institute (DLI) Course Fundamentals of Deep Learning, along with Next Steps project.

Labels

Computer Vision DLI Jupyter Machine Learning Machine Learning & AI

Pull Command

```
docker pull nvcr.io/nvidia/dli-dl-fundamentals:v0.0.1
```

1단계 Docker 설정

<https://www.docker.com/>

2단계 NGC 카탈로그 방문

<https://ngc.nvidia.com/catalog/containers/nvidia:dlf-fundamentals>

3단계 컨테이너 Pull 및 Run

[localhost:8888](#)



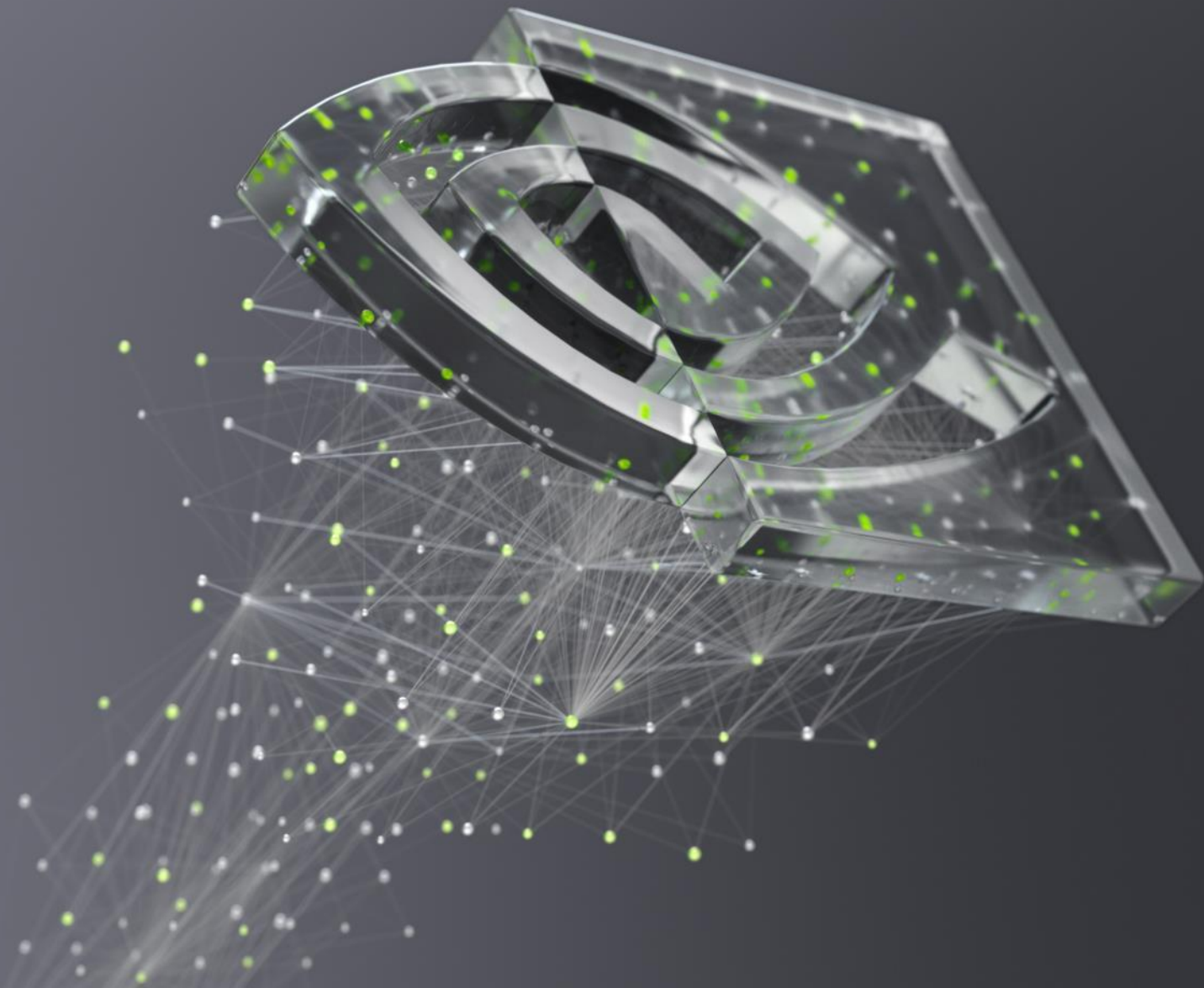
결론

로켓 사이언스 모방하기





시작하겠습니다!



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